



The Impact of Mobile Broadband and Internet Bandwidth on Human Development—A Comparative Analysis of Developing and Developed Countries

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Received: 7 May 2022 / Accepted: 12 December 2023 / Published online: 26 January 2024
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Abstract

This paper examines the impact of information and communication technologies (ICT) on human development in developing and developed countries, measured by the human development index (HDI). The analysis relies on new and contemporary measures of ICT, namely mobile broadband and internet bandwidth, which have only recently become available for many countries. Using data from 180 sample countries over the period 2010–2017, the system GMM estimates suggest that the impact of ICT on human development depends on the country's development stage and the respective telecommunication service. Mobile broadband drives human development in developing countries, while developed countries gain from increasing internet bandwidth. Further analysis reveals that the positive effects in developing countries are due to improvements in health and education. In contrast, in developed countries, this progress is attributable to positive effects on income.

Keywords Mobile broadband · Internet bandwidth · ICT · Human development · Panel regression

Introduction

Over the last three decades, significant advancements in information and communication technologies (ICT) have facilitated numerous innovations and the dissemination of knowledge. The sustained improvements in ICT, particularly mobile and fixed broadband telecommunication services, can contribute to better access to education and health services and increased efficiency in producing goods and services, thus affecting human development in its myriad forms (ITU, 2017). As the comprehensive literature reviews by Cardona et al. (2013) and Gómez-Barroso and

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Marbán-Flores (2020) indicate, most empirical studies confirm a positive connection between ICT and economic growth.

While economic growth is an important aspect, it is not the only impact of ICT. Recognising the limitations of focusing solely on macroeconomic aggregates (e.g. GDP), there is a growing consensus that investigating the influence of ICT on development as a more comprehensive concept is essential (e.g. Thapa & Sæbø, 2014; Heeks, 2010; Kleine, 2009). One of the most influential theories advocating a broader perspective on development is Amartya Sen's capability approach (Sen, 1999). This approach centres around people's capabilities or functioning, emphasising what individuals can effectively do or become. While economic growth undoubtedly contributes to the capability of 'being able to afford a decent life', the capability approach recognises that it is not the sole determinant of development. To foster the capability of individuals to live a life they value, additional dimensions such as education and health are crucial. Building on this framework, the United Nations Development Program (UNDP) introduced the concept of human development in 1990, offering a more comprehensive perspective. As a measure of human development, the UNDP proposed the Human Development Index (HDI), which encompasses dimensions such as income, health, and education. The HDI has been an essential metric in numerous empirical studies, including the one presented here.

Technology plays a crucial role in the various attributes that affect human development. Oosterlaken (2015) points out that the capability approach is highly compatible with the use of technology, as one of the primary functions of technology is, or should be, to enhance the capabilities of users. In this context, scholars have proposed models explaining the role of ICT in human development using the capability approach (e.g. Zelenkov & Lashkevich, 2022; De La Hoz-Rosales et al., 2019). Our study builds upon this theory, positing that increased access to ICT can significantly expand human capabilities by improving access to timely information and connectivity. Capability scholars also emphasise the contextual differences that influence the relationship between technology and capability expansion, including differences in socio-economic status, education, and infrastructure (Leach & Scoones, 2006). For instance, access to technology may not necessarily lead to increased capabilities if other barriers exist, such as affordability, availability, and digital literacy. Acknowledging these contextual differences, this study seeks to advance the understanding between specific aspects of technology adoption, including technological (mobile versus fixed broadband), type of telecommunication service (basic mobile phone subscriptions versus mobile broadband subscriptions and fixed broadband subscriptions versus broadband speed), as well as country characteristics (developing versus developed countries).

Building upon the concept of technological differences, researchers have utilised various ICT measures to investigate the relationship between ICT and development. The technical development of mobile phone and fixed broadband technologies has closely linked these measures. Until the 2000s, fixed-line telephone penetration was a relevant indicator to measure the ability to communicate over distances. However, mobile phone penetration has gradually replaced it in recent years. This paper uses active mobile broadband subscriptions per 100 inhabitants (mobile broadband) as a proxy for the mobile telecommunication service. This selection is driven by the

fact that mobile broadband contributes significantly to the improved usage quality of mobile devices and partly substitutes traditional mobile telecommunication services, such as mobile cellular subscriptions per 100 inhabitants (mobile basic). While mobile devices in the 1980s only allowed telephoning or sending an SMS, data transport rates of at least 256 kbit/s are now possible in many regions. Therefore, mobile broadband provides individuals with the capability to use advanced applications such as audio and video transmission, online banking, and web browsing and replaces the traditional core functions of mobile devices. A growing share of data traffic from mobile devices is shifting to mobile broadband-based phone and chat functions, avoiding the charges for call and SMS usage. Due to this substitution mechanism, increased mobile broadband penetration leads to decreased demand and relevance for mobile basic. The choice of mobile broadband is reinforced by the growing research interest (see Edquist et al., 2018) and its rapid deployment in recent years. While there was no significant mobile broadband penetration in the early 2000s, subscriptions rose to over 5 billion by 2020 (Ericsson, 2021).¹ Nevertheless, this study also considers the traditional indicator mobile basic for comparability with existing studies. Mobile basic includes the number of postpaid subscriptions and active prepaid accounts that enable voice communications, SMS, and data communications at low and medium speeds.

Concerning fixed broadband technology, in 2001, Norris described the internet as the ‘medium of choice par excellence’ (Norris, 2001, p.24), referring to the open nature of the internet and its potential for expanding human capabilities. However, more than mere internet access is required to exploit its opportunities fully. Many internet-related technologies and applications rely on adequate connection speeds (see California Broadband Task Force (2008) for an overview). To address this requirement in this study, we utilise international bandwidth usage (internet bandwidth) as a proxy for fixed broadband telecommunication services. This measure provides information about the average data transmission rate from one country to other parts of the world. Due to increasing global interconnectedness and the requirements for progressively higher data transmission rates for internet applications, the availability of high internet bandwidth is a prerequisite to realise the full potential of internet access. In contrast to fixed broadband subscriptions per 100 inhabitants (broadband basic), which provides information about the general capability to access the internet, internet bandwidth gives valuable insights into the impact of quality of use and speed. An in-depth analysis in this direction is highly relevant and necessary for an accurate conclusion, as even if countries have the same amount of fixed broadband subscriptions, differential utilisation possibilities due to internet bandwidth can be assumed (ITU, 2012). Similar to mobile broadband, internet bandwidth has also gained considerably in importance—both for the population and researchers (see Gruber et al., 2014). Technological developments, in particular the emergence of next-generation networks, have led to a surge in the total used capacity of international connections in recent years, as the increase of the

¹ Most subscriptions are for smartphones, whereas mobile PC and tablet subscriptions play only a minor role.

worldwide international internet bandwidth from around 600 Gbit/s in 2001 to 252 575 Gbit/s in 2017 clearly shows (ITU, 2014; ITU, 2018). Additionally, broadband basic, which refers to fixed subscriptions to high-speed access to the public internet that provides downstream speeds equal to or greater than 256 kbit/s, is also used in the analysis.

A priori, the impact of ICT on human development is not clear, as it is expected that the effects depend on a country's level of development. On the one hand, developing countries might benefit from leapfrogging older technologies and other latecomer advantages, such as the falling hardware price and the ability to imitate superior technologies (Fong, 2009; Blomstrom & Wolff, 1989). On the other hand, the disadvantages of being a latecomer, such as lacking complementarity factors and absorptive capacities, potentially surpass the advantages (Keller, 2004). Whether and which aspect prevails could lie in the nature of the respective telecommunication service. While with mobile telecommunications, developing countries have the potential to leapfrog the older technology based on fixed telephone lines, with fixed broadband, no earlier version of a similar technology can be leapfrogged.

The increasing ICT penetration rate observed over the years has attracted the attention of both researchers and policymakers to its impacts on human development. Thus, scholars have examined the impact of ICT indices or single telecommunication services, such as the number of fixed telephones, mobile phones, and fixed broadband subscriptions. The trend in the results is that ICT has a positive effect on human development (e.g. Nipo et al., 2023; Karaman Aksentijević et al., 2021; Njoh, 2018; Lee et al., 2017; Asongu & Nwachukwu, 2016). However, an extension of the existing studies is necessary as they underly three shortcomings. First, their analyses do not account for the fast-changing technological environment. Technological change is accelerating, with innovations and applications emerging at an unprecedented rate. This rapid evolution can dramatically alter the impact of ICT on human development, rendering findings from studies that do not account for these changes potentially outdated or incomplete. In Gómez-Barroso and Marbán-Flores (2020, p. 10) words, 'There is an inevitable lag between network growth and innovation on the one hand, and research on the other. Second, past macro-level studies primarily provide relevant findings only in the context of small samples of developing and developed countries, region-specific investigations, and country-specific investigations. However, a simple but interesting question remains insufficiently addressed: Which countries—developed or developing—tend to benefit from which telecommunication service? Third, past studies using the HDI as a proxy for human development mostly ignore the channels through which ICT influences the overall index. Changes in the overall HDI can only tell part of the story. The conclusion may be incomplete without analysing which components are responsible for the change.

This study aims to fill these gaps. Specifically, the main objective is to empirically investigate the relationship between increasing ICT penetration and human development, considering different technology and country characteristics. To contribute to the existing literature, we address the prevailing shortcomings in three ways. First, we use panel data of 180 countries divided into two groups resulting in 121 developing and 59 developed countries, covering more than previous studies.

Second, we are the first to use mobile broadband and internet bandwidth as a proxy for ICT. Combined, these two specifications allow us to directly identify any differential impacts of the respective technologies and recommend more targeted policy measures for the respective group of countries. Furthermore, this study extends the analysis to the HDI dimensions of income, health, and education individually. This approach provides unique insights into the channels through which telecommunications services influence human development.

Findings suggest that mobile broadband catalyses human development in developing countries, while internet bandwidth has no statistically significant effect. In developed countries, the positive and statistically significant effects result solely from increasing internet bandwidth, while mobile broadband does not contribute to changes in the HDI. The traditional mobile and broadband basic measures are statistically insignificant in both country groups. Regarding transmission channels, ICT does not affect income in developing countries; instead, human development gains are channelled through improvements in health and education outcomes. In developed countries, the positive effect is exclusively attributable to the income dimension.

The rest of the paper is organised as follows: the “Literature Review” section reviews the literature pertinent to the topic, focusing on the technological distinctions between mobile and fixed broadband telecommunications and the three dimensions of the HDI: income, health, and education. The “Empirical Analysis” section describes the empirical model and the unique features of the data set used. The “Empirical Results” section presents the findings of the empirical analysis. The “Discussion and Conclusions” section concludes.

Literature Review

Theoretical underpinnings

From a theoretical perspective, it is not straightforward to determine whether and how ICT affects human development. Two central factors, which require simultaneous consideration, decisively determine the outcome: Firstly, the country’s level of development. The so-called ‘latecomer advantage’ forms the basis of this concept. Gerschenkron (1962) developed the theory suggesting that developing countries have the potential to catch up with the mature technology of their developed counterparts. While the discussion in Gerschenkron was limited to the catching-up process of mature technology, Perez and Soete (1988) further developed the idea. They explain how latecomers leverage emerging technological paradigms as a window of opportunity to bypass investment or capacity-building stages that countries previously had to go through during development. This concept of ‘technological leapfrogging’ can result in several advantages for latecomers. They can better leverage emerging technologies because they rely less on existing technologies and systems than more advanced economies. Furthermore, they can bypass incomplete technologies in the initial stages of development, which entail high costs due to R&D investments.

In more recent literature, there is a consensus that ICT could provide significant avenues for technological leapfrogging for developing countries in the sense mentioned above (e.g. Lee et al., 2021; Fong, 2009; Steinmueller, 2001). However, the existing literature also indicates that the prerequisite for technological leapfrogging is the existence of appropriate absorptive capacities. Absorptive capacity is the ability of individuals and organisations within a country to comprehend, assimilate, and effectively utilise new technologies (Keller, 2004). This includes the ability to acquire knowledge and skills related to the technology, as well as the ability to adapt existing systems and processes to incorporate the new technology. Absorptive capacities are influenced by factors such as education levels, research and development capabilities, and access to ICT. Pronounced absorptive capacities are essential for countries looking to leapfrog technological development stages and adopt advanced technologies.

To fully comprehend the distinct effects of ICT on human development, considering the countries' level of development is crucial. Nevertheless, in this context, another relevant factor should be accounted for: The distinctive technological characteristics of mobile and fixed broadband telecommunication services (McDonough, 2016). This idea aligns with the capabilities approach basic assumption that the presence of resources does not always lead to expanding each person's capabilities, which applies to telecommunication services as well as to other technologies. There are compelling indications that mobile technology has the potential to be a leapfrog technology, which at the same time has a low restraint in terms of absorptive capacity (e.g. Aron, 2015; Napoli & Obar, 2014; James, 2013; Fong, 2009). Two main factors contribute to this specificity of mobile technology. First, the cost and complexity of integrating new electronic switching exchanges into an electromechanical infrastructure are much higher than building a network of entirely electronic switching from scratch (Antonelli, 1991). This has made it particularly beneficial for developing countries to rapidly establish a widespread mobile network, leapfrogging the more expensive and time-consuming process of building fixed-line telephone networks. By way of illustration, consider the African region. In 1999, only 10% of the population had access to a 2G network or better. The access increased to 60% by 2008 and 88.4% by 2020. Alongside the improved coverage, the population witnessed a significant increase in mobile phone subscriptions from 32.2% in 2008 to 82.4% in 2020. At the same time, the relevance of fixed telephony continued to decline, as evidenced by the decrease in fixed-line subscriptions from 1.5% in 2008 to 0.7% in 2020 (ITU, 2023).

Second, in developed countries, mobile phones and their related functionalities complement fixed-line telephony and existing services, rather than providing access to previously unavailable services (Waverman et al., 2005). In contrast, the diffusion of mobile phones introduces the potential to use services and applications in developing countries that were previously unavailable, either in digital or analogue forms, which is why different effects compared to developed countries are expected. Here, the substitutability of fixed-line telephony with mobile phones is not a like-for-like substitution but provides various additional functionalities (Heeks, 2010). Potential mechanisms through which mobile phones can spur development include improving access to and use of information, reducing search costs, increasing market efficiency, improving labour market outcomes, and generating additional job

opportunities (Aker & Mbiti, 2010). In this context, the mobile money sector is a prime example of leapfrogging in developing countries. Through expanding mobile phone access, financial and payment services for the unbanked population without access to offline banking can be provided (Lee et al., 2021). Most of the 866 million registered mobile money users in 2018 were from Sub-Saharan Africa, comprising 45.6% of the global total, followed by South Asia with 33.2%, and East Asia and the Pacific with 11% (GSMA, 2018). In most developed countries, on the other hand, mobile money plays a minor role, as traditional banking services are still the primary means of accessing financial services.

In contrast to mobile telecommunications, two significant factors suggest that the benefits of fixed broadband telecommunications are more likely to materialise in developed countries. First, fixed broadband technology does not allow bypassing any older version, rendering the development-promoting effects of the leapfrogging argument inapplicable to developing countries. It presents an improvement over older technologies, such as dial-up internet or copper-based broadband, rather than introducing a new, innovative technology that bypasses existing infrastructure. Furthermore, building up or improving fixed broadband networks requires significant investment in physical infrastructure such as cables, wires, and other physical infrastructure to deliver internet connectivity, which can limit expansion in developing countries (Kim et al., 2010). Although mobile networks can substitute fixed broadband in certain areas, especially in developing countries, in terms of speed and reliability, they are unlikely to be a complete replacement for next-generation broadband² (e.g. Napoli & Obar, 2014; Lee et al., 2011; Lewin et al., 2009). For developed countries, it can be precisely this speed and reliability improvement which help them to improve their already highly mature economic, health, and education systems. Such applications are high-definition video consultations, movement of terabyte datasets, and telemedicine remote control and require high-speed broadband connections that allow transmissions between 1 and 10 Gbps (California Broadband Task Force, 2008).

Second, using fixed broadband networks imposes considerably higher requirements on the necessary preconditions for citizens, businesses, and governments to successfully integrate it into achieving self or collective goals (Gurstein, 2003). Even if a country has widespread penetration and use of fixed broadband networks, there is little overall economic and social benefit if the absorptive capacity is limited (Kelly & Rossotto, 2011). In other words, in the case of two countries with the same level of development and fixed broadband penetration, differential effects due to differential levels of absorptive capacity are expected. In developing countries particularly, this could curb positive outcomes, as essential preconditions such as complementary R&D spending, country- and culture-specific web content, and adequate levels of human capital often not available to a sufficient extent. Especially with the latter, the use of the respective telecommunications differs enormously. While using mobile telecommunication services requires only a minimum of numeracy skills,

² Next-generation broadband generally refers to broadband services from the next-generation network (NGN) that can provide fast and ultra-fast internet services. It can be available from fixed and mobile networks but mainly refers to connections from fixed networks.

utilising fixed broadband telecommunications requires English language skills, computer skills, and technical competence. It can be assumed that these characteristics are more pronounced in developed countries, allowing them to reap greater benefits from fixed broadband technologies (James, 2013).

Based on the theoretical elaboration discussed above, it becomes apparent that the impact of telecommunications on human development is a complex and multifaceted phenomenon that depends on various factors. One of these factors is the country's level of development, which influences its ability to adapt and absorb new technologies. Here, the specific technical characteristics of the ICT infrastructure play a crucial role in determining their impact on human development. Building on this understanding, we hypothesise that the positive impact of mobile telecommunications on human development is more pronounced in developing countries. In contrast, developed countries experience more benefits from the impact of fixed broadband telecommunications.

Empirical Literature

Although telecommunications have been present for many years, macro-level empirical work on its impacts on human development is a relatively new area of exploration. In this section, we focus on three strands of literature that are close to this study: studies that consider the country's level of development when investigating the ICT–human development nexus, studies that investigate the impact of specific telecommunication services on human development, and studies that examine telecommunications impact on the single HDI components.

Addressing our research objective to examine the impact of telecommunications on human development in developing and developed countries, one strand of literature explores the global level. Specifically, it investigates the relationship between ICT and human development, considering the influence of the country's level of development. Karaman Aksentijević et al. (2021) examine the impact of an ICT index on the HDI during 2007–2019 for a panel of 130 countries. The results suggest a significant positive effect in lower-middle-income and low-income countries, while the effect is insignificant in high- and middle-income countries. As a possible reason for these disparities among the different income groups, the authors state that developed countries might have reached a peak in ICT penetration and that technological innovations such as artificial intelligence and cognitive technologies could be more important for them than mere ICT use. Deviating from these results, De La Hoz-Rosales et al. (2019) show in a similar research setting that ICT, in their study represented by 'individual use of ICT', has significant positive effects on human development regardless of the level of development. However, proxying ICT with 'government use of ICT' or 'ICTs for commercial purposes' leads to different results in the respective country groups. Ježić et al. (2022) study for the period from 2007–2019 examines the nexus between ICT, represented by an ICT use index, and the HDI for 130 countries divided into four income groups. The FE estimator indicates that ICT has a significant, positive effect across all countries examined. In contrast, employing the GMM estimator and thus containing potential endogeneity

concerns, a significant impact of ICT can only be found in upper-middle-income countries. Zelenkov and Lashkevich (2022), using a path modelling technique and a sample of 115 countries for 2019, find differences in how ICT influences the countries of the different country groups. In developing countries, ICT contributes to human development at the individual level only, while in developed countries, ICT contributes more to social conversion factors and less to personal ones. Lee et al. (2017), who employ a year lagged seemingly unrelated regression analysis with a sample of 102 countries from 2000–2013, investigate how telecommunication services affect human progress in different country groups. Among others, they present two relevant findings. First, in a common sample of low-income, lower-middle-income, upper-middle-income, and high-income countries, the impact of broadband, mobile phone, and fixed telephone penetration on human progress is positive and significant. Second, splitting the sample by country groups, broadband penetration is only positive and statistically significant in the high-income group and insignificant (and negative) in the other income groups. Regarding mobile phone penetration, results indicate a significantly negative impact on human progress in high-income countries. Contrary, in upper-middle-income and partly in lower-middle-income countries, the impact is positive and significant. In low-income countries, the effect is insignificant.

As this literature review shows, all studies use exclusively ICT indices for analysis, except for Lee et al. (2017), who also consider individual telecommunication services. Furthermore, the studies used different samples, periods, and measures, limiting the results' comparability and generalizability. However, some common findings provide a starting point for addressing our underlying research questions. Firstly, ICT has a positive impact on human development in most cases. Second, the impact depends on a country's level of development, while there is no clear trend as to which group benefits more.

To fulfil our research objective of investigating the differential impacts of mobile and fixed broadband telecommunications on human development in developing and developed countries, it is essential to explore the existing literature that examines the impact of individual telecommunication services. However, it is worth noting that these studies often have a narrow focus on specific regions rather than considering a global perspective. For example, Iqbal et al. (2019), using data from five South Asian countries from 1990 to 2016, find a positive and significant effect of mobile phone penetration on human development. In contrast, the coefficient of internet penetration is insignificant. Using a similar country group for a shorter period (2000–2016), Gupta et al. (2019) find that all ICT measures, internet, broadband, and mobile phone penetration, have a statistically significant positive effect on the HDI. The deviating results of the two studies could stem from the different periods because, before 2000, the internet had virtually no relevance in the region. Another study that examines the Asian region, more specifically 46 countries from 2010–2019, is Nipo et al. (2023). Their findings reveal that internet usage significantly correlates with the HDI for Asia-Pacific countries with high and medium human development. Mobile cellular subscriptions emerged as significant only for these Asian countries with high human development. Asongu and Nwachukw (2016) and Njoh (2018) using a sample of SSA countries for the period 2000–2012 and of

African countries for 2013, respectively, find evidence that mobile phone penetration and internet access foster (inclusive) human development. However, the latter do not find such evidence for fixed telephone and broadband penetration, illustrating that the respective telecommunication services have different effects on human development. According to the author, the lack of statistically significant effect is not surprising, given that possible positive effects are negated by diseconomies of scale, among other things, which result from the low penetration rates of fixed telephone and fixed broadband in the region. In the context of transition economies, Bayar et al. (2023) present a comprehensive study on 11 transition economies of the EU for the period 1996–2021 and reveals that ICT penetration, represented by the use of the internet and mobile cellular subscriptions, has a significant influence on human development in both the short and long term. Considering a sample of 28 EU member states for 2011–2017, Petrić et al. (2020) find mixed results when examining the influences of the individual components of the Digital Economy and Society Index on the HDI. Their findings suggest that broadband internet, internet usage per individual, and the use of e-commerce in enterprises have significant positive effects on the HDI. In contrast, the measures of ICT professionals employed domestically and individuals using government services have the opposite effect.

Similar to global studies, research focusing on specific regions tends to find a positive relationship between ICT and human development. In addition, findings reveal that different telecommunication services tend to affect human development differently. However, there is no clear trend in how the impact of mobile and broadband telecommunications differs. Again, different analytical approaches and data sets covering different countries and periods might explain this.

The third strand of literature, including studies that have explored the nexus between telecommunications and income, health, and education, is reviewed to address our research objective to investigate telecommunications' impact on the single HDI components. Concerning the income dimension, in a recent literature review, Gómez-Barroso and Marbán-Flores (2020) find that most econometric studies firmly establish the significant role of ICT in influencing countries' economic development. However, the results become less conclusive when examining different country groups and individual telecommunication services separately. Focusing on mobile phone penetration, Waverman et al. (2005), using a sample of 92 high-income and low-income countries from 1980–2003, reveal that the positive impact of mobile phones may be twice as large in developing countries compared to their developed counterparts. The authors relate these results to the different preconditions of fixed-line penetration in the country groups. Due to a relatively low fixed-line penetration in developing countries, mobile phones are a substitute. Thus, they can achieve more significant growth effects than their developed counterparts—a finding that aligns with more recent investigations (e.g. Kumari & Singh, 2023). Bahrini and Qaffas (2019) investigated the impacts of ICT on economic growth in 45 selected developing countries in the Middle East and North Africa (MENA) and Sub-Saharan Africa (SSA) from 2007–2016. Their results suggest a different impact of the respective telecommunication service, depending on the country's level of development (in the sample used, the mean GDP per capita is almost seven times higher in MENA than in SSA). On the one hand, results regarding

mobile phone penetration indicate a positive impact on economic growth in both regions, but more substantial (+2.4%) and statistically significant in SSA. On the other hand, the impact of internet usage and broadband penetration is more substantial in MENA than in SSA (+2.7% and +2.23%, respectively). Similar results are provided by Myovella et al. (2020) using data from 41 SSA countries and 33 OECD countries from 2006–2016. This general direction cannot be confirmed by all empirical studies, as also dissenting evidence is provided for a more substantial positive impact of mobile phone penetration on GDP in high-income countries in the 1990–2007 period (Gruber & Koutroumpis, 2011); a more substantial positive impact of fixed broadband penetration on GDP growth in developing countries in the period 1980–2006 (Qiang et al., 2009); and a more substantial positive impact from mobile phone penetration than from internet penetration in 28 EU countries from 2000–2017 (Toader et al., 2018).

Turning to the health dimension, studies by Tariq Majeed et al. (2019) and Lee et al. (2016) present evidence of a positive and significant relationship between ICT indices and health in a large sample comprising both developing and developed countries. However, more disaggregated studies suggest that the effects may vary depending on the level of development. Several studies conducted in the African region have confirmed a positive relationship between ICT and health. For instance, Kouton et al. (2021), Adeola and Evans (2018), and Mimbi and Bankole (2015) find evidence supporting the positive impact of ICT on health outcomes. Notably, Mimbi and Bankole (2015) observe that the main telephone line had the highest return on health outcomes among the ICT components investigated, followed by internet usage and mobile phone penetration. Contrary, Rana et al. (2018), using a sample of 30 OECD countries for the period 2004–2015, suggest a significantly negative effect of ICT access and use on health outcomes. Although the few studies available give a first impression about ICTs' effects on health, more robust evidence is needed.

Regarding the educational dimension, the least empirical macro-level evidence is available. Asongu and Nwachukwu (2018) and Asongu and Odhiambo (2019) examined a sample of 49 Sub-Saharan African countries from 2000–2012. Their findings highlight the positive effects of mobile phones on educational quality and inclusive development. Specifically, they find that increasing mobile phone and internet penetration is associated with improving primary education quality. Asongu and Odhiambo (2019) point out, among other reasons, the potential of using ICT in the classroom to improve the situation characterised by overcrowded classrooms, a lack of adequate teaching materials, and crumbling infrastructure. To investigate the impact of ICT on the education index, Oyerinde and Bankole (2019) employ a Data Envelopment Analysis and use data for Africa and selected countries in Europe and Northern America for the period 2010–2016. The main finding is that ICT positively impacts educational attainment and adult literacy rates. Furthermore, the study also shows that regions with significantly lower education and literacy rates use their less developed ICT infrastructure relatively efficiently, providing evidence for confirming the leapfrogging hypothesis in this context. The survey of this strand of literature reveals that there is a clear tendency for ICTs to have a positive impact on the individual HDI components. However, this is not without limitations, as there is also contrary evidence, and the results differ depending on the sample and the

indicator used. Particularly in the areas of health and education, more studies need to be carried out to draw a robust conclusion about the interrelationships.

Based on the above literature, one can draw the following conclusions: ICT generally benefits human development in developing and developed countries, and the impact mostly depends on the telecommunication service. However, so far, studies have not provided robust empirical evidence to determine which telecommunication service (mobile or fixed telecommunications) affects the development of which group (developing or developed countries). This study aims to address this question in detail. It stands out from previous literature by using mobile broadband and internet bandwidth as proxies for ICT, indicators that have been previously overlooked in this context. Furthermore, this study not only analyses the impact on the HDI but also considers the individual components separately.

Empirical Analysis

Model

The econometric analysis aims to test the hypothesis that (i) human development in developing countries is positively affected by mobile telecommunications and (ii) human development in developed countries is positively affected by fixed broadband telecommunications. For this study, a dynamic panel data model is used, taking the following form:

$$HDI_{it} = HDI_{it-1} + \beta_1 ICT_{it} + \beta_2 X_{it} + \lambda_t + \mu_i + \varepsilon_{it} \quad (1)$$

where the subscripts refer to country i and year t . HDI, representing human development, enters the model as the dependent variable. Lagged HDI (HDI_{it-1}) is included to control for autocorrelation and omitted variables. It is assumed that HDI depends on the vectors of variables ICT, representing fixed broadband subscriptions, internet bandwidth, mobile phone subscriptions, and mobile broadband subscriptions. X is the vector of the control variables FDI, inflation, GDP growth, population growth, urban population growth, and corruption. Furthermore, μ represents unobserved specific terms for each country, and λ denotes time dummies. The final term, ε , is the random error term.

The relationship between the ICT variables and human development can be influenced by endogeneity and reverse causality, as the literature suggests that technology expands human capabilities but also that human capabilities shape technology to a certain extent (Oosterlaken, 2015). Thus, the OLS estimator could yield upward biased estimates, and the Within Group estimator is also unreliable as it may generate downward biased estimates (Nickell, 1981). To address these issues, Anderson and Hsiao (1982) propose a first-differenced transformation to eliminate fixed effects. However, this transformation does not eliminate the correlation between the differenced error term and the differenced endogenous regressors. To mitigate these problems, a difference generalised method of moments (GMM) estimator is suggested. In this approach, the differenced endogenous explanatory variables are instrumented

with suitable lags to ensure that the instruments are uncorrelated with the error term, as Arellano and Bond (1991) outlined. However, Arellano and Bover (1995) and Blundell and Bond (1998) point out that using the lagged level of endogenous variables as instruments for the first-differenced variables may be ineffective. As an alternative, they propose the system GMM estimator. The basic idea of the system GMM estimator is to estimate the empirical model as a two-equation system, which offers the possibility to combine differences and levels of variables, addressing the limitations of using solely differences. This involves utilising lagged differences instruments for levels and lagged levels of instruments for differences, effectively capturing both short-term dynamics and long-term relationships in the data. One ideally would tackle endogeneity concerns using external instruments, but pure exogenous instruments that vary across countries and over time are rarely found. While these concerns are challenging to tackle with traditional methods, the system GMM utilises internal instruments to control for endogeneity across all explanatory variables. In addition, the system GMM offers the advantages that cross-country variations are not eliminated and that it is more efficient than the difference GMM in cases where the variables are persistent.

Due to these considerations, the chosen approach for estimating Eq. (1) is the system GMM method, as proposed by Arellano and Bover (1995). This study prefers the two-step system GMM over the one-step system GMM. Although this provides more efficient estimates (Blundell & Bond, 1998), the associated asymptotic standard errors may be downward biased in finite samples. To adjust for potential biases when using the two-step system GMM, the Windmeijer finite-sample correction for the linear efficient two-step system GMM estimator variance is applied (Windmeijer, 2005). A further specification is the orthogonal-deviations transformation technique, which removes the fixed effects (the time-invariant country characteristics correlated with the regressors). This allows as many observations as possible to be included in unbalanced panels (Arellano & Bover, 1995). All explanatory variables are treated as suspected endogenous, while only time dummies are treated to be strictly exogenous.

While the GMM estimator has several advantages, such as its ability to handle endogeneity concerns, it has been criticised for not being robust to the choice of instruments, especially in larger models where weak instruments can lead to biased estimates.³ To ensure its accuracy, the validity of two underlying assumptions that underlie this estimator must be tested (Roodman, 2009). First, we evaluate the validity of the instruments by utilising the Hansen test, which tests the joint validity of the instruments. The second assumption is the absence of autocorrelation in the error terms, which we test using a second-order autocorrelation test (by construction, the differenced error term is probably first-order serially correlated even if the original error term is not). In addition to these tests, we guard against overfitting bias, which can arise if the endogenous variables' lags exceed the number of countries. To mitigate this risk, we limit the number of lags of the endogenous variables to the second to fifth lag, and the instruments are used in a collapsed matrix format.

³ See Roodman (2009) and Bazzi and Clemens (2013) for a comprehensive critique of GMM estimators.

Using a collapsed matrix format limits the number of moment conditions by writing all instruments related to a given observation of the dependent variable in one column, rather than having multiple columns for each period.

Data and Descriptive Statistics

This study prioritises obtaining a representative sample that includes as many countries as possible, with particular consideration given to the availability of ICT variables. Before 2010, in many countries no data was collected on mobile broadband connections, limiting data available for those years. For the period after 2017, international bandwidth data is often unavailable in developing countries, further limiting the pool of available data. This results in an unbalanced panel data set of 180 countries, 121 developing countries, and 59 developed countries, from 2010–2017. The classification of countries is based on the UNDP's cut-off points for grouping countries (UNDP, 2015). Countries with very high HDI in 2017 represent developed countries, and countries with high, medium, and low HDI represent developing countries. See Table 8 in the Appendix for the list of countries and their classification.

The measure for human development is the human development index (HDI) which is retrieved from the United Nations Development Program (UNDP).⁴ This index incorporates the dimensions: life expectancy at birth (health), expected years of schooling and mean years of schooling (education), and GNI per capita measured in PPP \$ (income). The performance of each of the three indicators is expressed as a value between zero and one. The average of the indicators represents the index, zero being the lowest and one the highest possible level of human development. The HDI is a recognised proxy for studying human development at the country level (UNDP, 2015).

Regarding the four ICT variables, data are obtained from the International Telecommunication Union (ITU).⁵ A detailed description can be found in the introduction, which is why we only briefly describe the data and their sources here.⁶ As a proxy for mobile telecommunication services, we use active mobile broadband subscriptions per 100 inhabitants (mobile broadband). This indicator pertains to the total number of active handset-based and computer-based (USB/dongles) mobile broadband subscriptions that allow access to the internet. It is limited to the count of active subscribers and does not include those who can subscribe but have not yet done so, despite having broadband-enabled devices. In this context, subscriptions need to consist of either a recurring fee or, in the case of prepayment, fulfil a usage condition: users must have connected to the internet within the past three months. For a mobile data subscription to qualify as broadband, it must promise the capability of using at least a 3G/UMTS network, so that a minimum download speed of 256 kbit/s

⁴ Available at: <http://hdr.undp.org/en/content/human-development-index-hdi>. Last accessed 24 June 2023.

⁵ Available at: https://www.itu.int/net4/itu-d/icteye#/. Last accessed 24 June 2023.

⁶ For a more detailed description of definitions, explanations and scope, collection method, relationship with other indicators, and methodological issues of the ICT variable, see the Handbook for the collection of administrative data on telecommunications/ICT (ITU, 2020).

can be expected. Subscriptions that solely utilise GPRS and EDGE technologies do not meet this requirement and are therefore excluded. The data collection can stem from licensed national mobile operators providing internet-accessible mobile broadband services and is then aggregated at the country level (ITU, 2020).

The traditional measure of mobile cellular subscriptions per 100 inhabitants (mobile basic) is also considered for comparability with existing studies. Mobile basic includes the number of postpaid subscriptions and active prepaid accounts that enable voice communications, SMS, and data communications at low and medium speeds. In addition to the description of the mobile telecommunication variables, one important limitation is worth considering. The indicators used for measuring telecommunication services consider subscribers and not users, which neglects the possibility of sharing. Especially in developing countries, there is a tendency to share tools for communication, driven by factors such as cultural norms and financial constraints (James, 2011). Therefore, the number of mobile phone users could be higher than what is being captured by subscription-based indicators. Thus, more precise measures could consider the access dimension, which refers to people who potentially use a mobile phone as they are within the coverage of a mobile network. Another option is to use the total number of mobile phone users, including those who regularly use mobile phones through ownership or sharing. However, such data is not readily available at the country level, which is why subscription-based indicators are currently the most appropriate proxies and are widely used in similar research settings (Edquist et al., 2018; James & Versteeg, 2007).

As for the proxy for fixed telecommunication services, international bandwidth usage in Tbit/s⁷ (internet bandwidth) is used in this study. It refers to the total used capacity of international bandwidth and is measured as the sum of the used capacity of all internet exchanges (locations where internet traffic is exchanged) offering international bandwidth. This can include optical fibre cables, radio links, and traffic processed by satellite ground stations and teleports to orbital satellites. Data providers are required to calculate the average over 12 months in the reference year and, in the case of asymmetric traffic (i.e. different incoming and outgoing traffic), to provide the highest of the two values. As far as data collection is concerned, there are several options. Data can be collected from facility-based carriers that provide wholesale international connectivity, from all operators in the country that contract or self-supply international bandwidth, namely fixed, mobile, and satellite operators, or other entities that may have direct connections to international carriers, namely over-the-top providers and content providers (ITU, 2020). Additionally, fixed broadband subscriptions per 100 inhabitants (broadband basic), which refers to fixed subscriptions to high-speed access to the public internet that provides downstream speeds equal to or greater than 256 kbit/s, is used in this study. The use of international bandwidth as a proxy for fixed telecommunication services in this study has some limitations. It measures the total used capacity of international bandwidth from one country to other parts of the world. However, it might not accurately reflect

⁷ The original variable is expressed in Mbit/s (megabits per second). The variable is transformed in Tbit/s (terabits per second) for better readability in the analysis.

the actual internet speeds experienced by users in that country, which is affected by other factors such as network infrastructure. While an alternative measure could be the average download and upload speeds experienced by users in a given country, such data is not available at the country level. Despite this limitation, international bandwidth usage provides valuable information for this study about the speed and quality with which a country's population can use the internet and other international data connections and is utilised in related research (ITU, 2018; Gruber et al., 2014).

In terms of incorporating additional controls into the model, no universally agreed-upon theoretical or empirical model defines the determinants of human development (Sen, 1999). However, economic prosperity and a stable macroeconomic environment have been identified to be significant predictors of human development (Suri et al., 2011). We address this in our model by including GDP annual growth rate in constant 2010 US\$ (GDP growth)⁸, GDP deflator as an annual percentage (inflation)⁹, and net inflows of foreign direct investment in percentage of GDP (FDI).¹⁰ Furthermore, demographic factors have also been popularised in the literature as potential determinants of human development (Todaro & Smith, 2012). To consider the impact of the size of a population over time and geographical location, we include the total annual population growth rate (pop. growth)¹¹ and the urban annual population growth rate (urban pop. growth)¹² in our model. Finally, functioning institutions can be a key determinant of human development (Shuaibu, 2016). In particular, corruption arises from fundamental institutional flaws that result in ineffective economic, social, and political consequences (Akçay, 2006). In our study, we consider this determinant by including the control of corruption indicator (corruption)¹³. This indicator ranges from -2.5 to $+2.5$, where a more negative or lower score indicates a greater level of corruption.

Table 1 presents the descriptive statistics for the variables used in the baseline estimations. The upper and lower panels show the summary statistics for the samples of developing and developed countries, respectively. As the HDI defines the split of the country samples, a considerably higher average HDI of 0.86 is seen in developed countries, compared to 0.62 in developing countries. Differences between the two samples also occur concerning the ICT variables, but the particular telecommunication service largely determines the magnitude. The number of mobile broadband subscriptions in developing countries is on average more than three times

⁸ Available at: <https://data.worldbank.org/indicator/NY.GDP.MKTP.KD.ZG>. Last accessed 24 June 2023.

⁹ Available at: <https://data.worldbank.org/indicator/NY.GDP.DEFL.KD.ZG>. Last accessed 24 June 2023.

¹⁰ Available at: <https://data.worldbank.org/indicator/BX.KLT.DINV.WD.GD.ZS>. Last accessed 24 June 2023.

¹¹ Available at: <https://unctadstat.unctad.org/wds/>. Last accessed 24 June 2023.

¹² Available at: <https://unctadstat.unctad.org/wds/>. Last accessed 24 June 2023.

¹³ Available at: <https://info.worldbank.org/governance/wgi/>. Last accessed 24 June 2023.

Table 1 Descriptive statistics (2010–2017)

Variable	Obs.	Mean	Std. Dev.	Min	Max
<i>Developing countries sample (121 countries)</i>					
HDI	968	0.617563	0.1195802	0.319	0.799
Mobile broadband	924	21.42905	23.21559	0	133.718
Mobile basic	968	88.14845	37.50081	1.17389	197.201
Internet bandwidth	945	0.2483172	1.059323	0	21.391
Broadband basic	956	4.08827	5.631144	0	31.9825
GDP growth	953	3.905107	6.373774	−62.07592	123.1396
Pop. growth	962	1.691664	1.170296	−1.745365	6.559122
Urban pop. growth	962	2.557299	1.605293	−2.975315	6.985868
Inflation	953	5.847493	7.76461	−26.70028	52.98898
FDI	952	4.582132	7.470553	−37.15476	103.3374
Corruption	968	−0.5313051	0.6267519	−1.826384	1.568311
<i>Developed countries sample (59 countries)</i>					
HDI	472	0.8622987	0.049574	0.743	0.953
Mobile broadband	465	68.51935	36.78296	0	241.197
Mobile basic	472	130.0406	27.00174	75.6288	251.765
Internet bandwidth	465	1.753064	4.976957	0.000032	53.5935
Broadband basic	472	24.5757	10.50172	1.30048	46.3058
GDP growth	472	2.590118	3.026847	−9.132494	25.16253
Pop. growth	472	0.8779413	1.514561	−1.854153	11.48337
Urban pop. growth	472	1.170688	1.617445	−2.282468	11.65055
Inflation	472	3.051004	7.51585	−25.95842	75.27737
FDI	472	8.654808	23.27655	−28.58479	280.1318
Corruption	472	0.9273414	0.8913106	−1.090531	2.404901

lower than in developed countries. Average mobile basic penetration is high in both samples, with developed countries having, on average, 130 mobile connections per 100 inhabitants, which is about 48% more than developing countries. Regarding the measures of fixed broadband telecommunications, the gap is considerably greater: Internet bandwidth is seven times higher in developed countries (1.75 Tbit/s) than in developing countries (0.25 Tbit/s), and in developed countries, there are 24.58 broadband subscriptions per 100 inhabitants on average compared to around 4.09 in developing countries.

Although the summary statistics show a relevant gap between developing and developed countries in all ICT measures, the growth curves shown in Fig. 1 indicate that this gap is narrowing. In developed countries, all variables except internet bandwidth point to saturation levels being reached. In particular, for mobile basic and broadband basic, the average growth rate for the whole period is below 0.1%, in the first case, even negative for 2017. In developing countries, the growth rate for mobile basic and broadband basic is significantly higher, but in developed countries, a continuous decline can be seen for both measures. Concerning internet bandwidth,

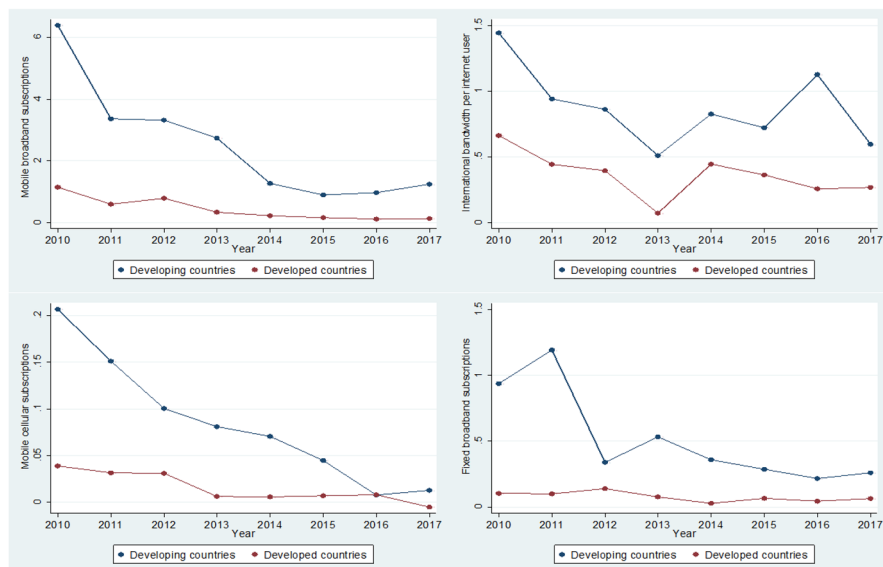


Fig. 1 Average growth rates of ICT telecommunication services in %

the annual growth rates vary relatively strongly, where no trend is discernible. Mobile broadband is the fastest-growing telecommunication service in developing countries, with an average annual growth of over 2.5%. Although the growth rate declined continuously between 2010 and 2015, a slight increase has been observed since 2015.

Empirical Results

Baseline Results

The baseline results of Eq. (1) estimation for developing and developed countries are reported in Tables 2 and 3, respectively. All regression models are estimated with the two-step system GMM estimator to address potential endogeneity problems, which satisfies several diagnostic tests. The Hansen test is applied to test the validity of the instruments. In all specifications, the validity of instruments can be confirmed. In addition, the AR(2) test is reported. No evidence for second-order serial correlation is found for all but four specifications. In cases where signs of second-order serial correlation are identified, the regressions are verified by adjustments in their specifications. After the adjustments, all equations perform well based on the diagnostic checks.

Regarding the control variables, their coefficients are as expected. In both samples, the coefficient on GDP growth is always statistically significant at the 10% level and higher in developing countries and the 1% level in developed countries,

Table 2 Baseline results for developing countries

	(1) Mobile broadband	(2) Mobile basic	(3) Internet bandwidth	(4) Broadband basic
HDI (t-1)	0.963109*** (0.014497)	0.951878*** (0.023515)	0.963328*** (0.018603)	0.955880*** (0.018034)
Mobile broadband	0.000074** (0.000035)			
Mobile basic		−0.000048 (0.000044)		
Internet bandwidth			0.000078 (0.000644)	
Broadband basic				0.000216 (0.000176)
GDP growth	0.000372* (0.000224)	0.000574*** (0.000093)	0.000591*** (0.000100)	0.000601*** (0.000083)
Pop. growth	−0.004452*** (0.001451)	−0.002573 (0.002915)	−0.004523*** (0.001696)	−0.005812*** (0.002024)
Urban pop. growth	0.001430 (0.001010)	0.000381 (0.001863)	0.002015 (0.001225)	0.002645* (0.001406)
Inflation	−0.000054 (0.000077)	0.000054 (0.000052)	−0.000039 (0.000060)	0.000006 (0.000040)
FDI	0.000055 (0.000142)	0.000063 (0.000097)	−0.000003 (0.000104)	0.000030 (0.000090)
Corruption	0.001373 (0.002380)	0.000783 (0.003879)	0.000544 (0.003324)	0.002484 (0.002608)
Constant	0.028778*** (0.009556)	0.040412** (0.017673)	0.027328** (0.012105)	0.033212*** (0.011921)
No of obs.	892	934	911	922
No. of countries	120	121	121	120
Instruments	48	48	48	48
AR(2) (p)	0.348	0.106	0.146	0.139
Hansen (p)	0.248	0.628	0.371	0.580

The dependent variable is the HDI. Robust standard errors in parentheses. All estimations include time dummies (coefficients not reported). The asterisks (*, **, and ***) denote 10%, 5%, and 1% significance levels, respectively

indicating that increasing income fosters human development. On the one hand, increasing GDP enables purchasing of more and better health and education services. On the other hand, a stable macroeconomic environment is a major prerequisite for the population's well-being. In contrast, population growth is generally statistically significant and negatively correlated with HDI in both samples. Especially in developing countries, population growth potentially negatively affects all three dimensions of human development. In terms of income, rapid population growth affects the economic capabilities of groups already poor and dependent on agriculture,

Table 3 Baseline results for developed countries

	(1) Mobile broadband	(2) Mobile basic	(3) Internet bandwidth	(4) Broadband basic
HDI (t-1)	0.925506*** (0.033903)	0.907198*** (0.041981)	0.968153*** (0.032983)	0.969497*** (0.045102)
Mobile broadband	−0.000039 (0.000026)			
Mobile basic		0.000001 (0.000035)		
Internet bandwidth			0.000099* (0.000054)	
Broadband basic				−0.000046 (0.000121)
GDP growth	0.000441*** (0.000147)	0.000513*** (0.000128)	0.000432*** (0.000147)	0.000556*** (0.000124)
Pop. growth	−0.001438** (0.000633)	−0.001740** (0.000722)	−0.001290* (0.000747)	−0.001111 (0.000931)
Urban pop. growth	0.001105 (0.000672)	0.001131 (0.000828)	0.001183 (0.000708)	0.000708 (0.000874)
Inflation	−0.000003 (0.000035)	0.000008 (0.000039)	0.000012 (0.000050)	0.000004 (0.000041)
FDI	−0.000026 (0.000018)	−0.000017 (0.000018)	−0.000016 (0.000012)	−0.000026 (0.000022)
Corruption	0.001920 (0.002343)	0.003484 (0.003155)	0.000104 (0.001845)	0.002061 (0.002599)
Constant	0.068503** (0.028391)	0.077935** (0.034351)	0.029640 (0.027150)	0.028181 (0.034607)
No of obs.	465	472	465	472
No. of countries	59	59	59	59
Instruments	48	48	48	48
AR(2) (p)	0.709	0.984	0.597	0.986
Hansen (p)	0.774	0.590	0.607	0.360

The dependent variable is the HDI. Robust standard errors are in parentheses. All estimations include time dummies (coefficients not reported). The asterisks (*, **, and ***) denote 10%, 5%, and 1% significance levels, respectively

particularly those with scarce land and natural resources. Furthermore, high fertility harms the health of mothers and children due to the increased risk of pregnancy. Finally, rapid population growth means a broader distribution of existing education expenditure, so the capability to obtain qualitative education is declining in favour of quantity (Todaro & Smith, 2012). The inflation, FDI, corruption, and in most cases, urban population growth coefficients are not statistically significant at conventional levels.

Concerning the main variables of interest, the estimates for mobile telecommunications in the 121 developing countries sample in Table 2 are considered. Mobile broadband is introduced in column 1, which is statistically significant at the 5% level. In contrast, the estimated coefficient on mobile basic, column 2, is statistically insignificant. These findings confirm the prior expectation that only mobile broadband is a relevant measure to explain changes in the HDI. Therefore, we exclude mobile basic from further analysis. Regarding the estimated coefficients on measures of fixed broadband telecommunications in columns 3 and 4, broadband basic and internet bandwidth are statistically insignificant.

From an economic perspective, column 1 results indicate that a one percentage point increase in mobile broadband leads, on average, to an increase in the HDI by 0.000074 points. To put this in perspective, if a country with an HDI of 0.500 were to increase its mobile broadband penetration by 10%, it could potentially increase its HDI to 0.50074. While this might seem like a minor increase, it could have significant implications for the country's global HDI ranking. Consider, for example, Libya in 2017. A 1% increase in mobile broadband would have led to an increase in HDI from 0.703536 to 0.70361. This increase, in turn, would have resulted in Libya overtaking South Africa in terms of HDI ranking.

Table 3 presents the results for the 59 developed countries' sample. In contrast to developing countries, none of the mobile telecommunication measures, shown in columns 1 and 2, are statistically significant, indicating that mobile telecommunications had no impact on the HDI in developed countries in 2010–2017. The situation is different for internet bandwidth. The estimated coefficient is statistically significant at the 5% level (column 3). Broadband basic, reported in column 4, has a statistically insignificant coefficient. According to these results, we consider internet bandwidth for further analysis as a representative measure for fixed broadband telecommunications.

The economic interpretation for developed countries suggests that a one percentage point increase in internet bandwidth leads, on average, to an increase in the HDI by 0.000099 points. In 2017, for example, Korea had a HDI of 0.903987. A 1% increase in internet bandwidth would have changed the HDI value to 0.904086, leading to significant implications for the country's global HDI ranking. In this illustration case, Korea would have overtaken Israel in the HDI ranking.

Robustness Checks

To ensure the validity of the baseline results, a series of robustness checks are undertaken, including a balanced panel, adjusting for outliers and variations in the GMM specifications.

So far, the models have been estimated using an unbalanced panel. Countries that lack data for a consistent time series may act as outliers, which would lead to biased results. We test whether the results are biased due to sample selection using an adjusted dataset containing only countries with observations on all elements over all periods. This approach reduces observations of about 26% for the estimations with mobile broadband, around 12% with internet bandwidth in the developing country sample, and

Table 4 Robustness I—Balanced panel and exclusion of potential outliers

Sample	(1) Developing countries	(2) Developing countries	(3) Developed countries	(4) Developed countries
	Winsorised at 1%			
HDI (t-1)	0.974535*** (0.012249)	0.988458*** (0.007993)	0.919803*** (0.045022)	0.939999*** (0.039428)
Mobile broadband	0.000080** (0.000033)		−0.000036 (0.000023)	
Internet bandwidth		−0.000780 (0.001571)		0.000126* (0.000070)
GDP growth	0.000543** (0.000270)	0.000383** (0.000184)	0.000546*** (0.000126)	0.000586*** (0.000085)
Pop. growth	−0.003852* (0.002033)	−0.005626*** (0.001463)	−0.001327 (0.000966)	−0.001392* (0.000810)
Urban pop. growth	0.000755 (0.001509)	0.002647** (0.001056)	0.000876 (0.001007)	0.000911 (0.000807)
Inflation	−0.000032 (0.000092)	0.000003 (0.000070)	−0.000002 (0.000045)	−0.000007 (0.000041)
FDI	0.000013 (0.000117)	0.000026 (0.000090)	−0.000024 (0.000024)	−0.000015 (0.000019)
Corruption	−0.002401 (0.001689)	−0.000782 (0.001744)	0.002900 (0.002890)	0.002335 (0.002697)
Constant	0.018432** (0.008138)	0.012345** (0.005677)	0.071598* (0.037294)	0.051700 (0.032224)
No of obs.	712	848	448	456
No. of countries	89	106	56	57
Instruments	48	48	48	48
AR(2) (p)	0.231	0.182	0.954	0.816
Hansen (p)	0.534	0.456	0.478	0.595

The dependent variable is the HDI. Robust standard errors in parentheses. All estimations include time dummies (coefficients not reported). The asterisks (*, **, and ***) denote 10%, 5%, and 1% significance levels, respectively

approximately 5% in the regressions with the developed countries sample. To test for other biases due to outliers, the HDI, mobile broadband, and internet bandwidth distribution for the respective sample is winsorised at the 1% level, as suggested by Barnett and Lewis (1994). Through winsorising, all values of the chosen variable at both distribution tails are replaced by a specified value.¹⁴ The estimated results of the telecommunications variables and GDP growth in Table 4 are robust and confirm the baseline results. More

¹⁴ For example, in the case of mobile broadband in the developed country sample, all values below the 1st percentile of mobile broadband distribution (1.56) are replaced with a value of 1.56. For mobile broadband values above the 99th percentile (155), all values are replaced with 155. This enables the usage of extreme observations without having a large impact on the estimates.

Table 5 Robustness II—Alternative estimator

Sample	(1) Developing countries	(2) Developing countries	(3) Developed countries	(4) Developed countries
HDI (t-1)	0.956263*** (0.017561)	0.965307*** (0.016962)	0.908419*** (0.040780)	0.932561*** (0.047186)
Mobile broadband	0.000105*** (0.000037)		-0.000052* (0.000029)	
Internet bandwidth		0.000246 (0.000640)		0.000109* (0.000058)
GDP growth	0.000354 (0.000259)	0.000417*** (0.000122)	0.000510*** (0.000150)	0.000562*** (0.000108)
Pop. growth	-0.005261*** (0.001560)	-0.004175* (0.002259)	-0.001310* (0.000779)	-0.001353*** (0.000500)
Urban pop. growth	0.001218 (0.001132)	0.002093 (0.001391)	0.000905 (0.000802)	0.000994 (0.000724)
Inflation	-0.000043 (0.000072)	-0.000042 (0.000055)	-0.000010 (0.000058)	-0.000004 (0.000045)
FDI	0.000072 (0.000148)	0.000042 (0.000089)	-0.000020 (0.000023)	-0.000010 (0.000010)
Corruption	-0.000129 (0.002779)	0.000551 (0.003796)	0.004028 (0.002911)	0.002764 (0.003029)
Constant	0.031485*** (0.011772)	0.025970** (0.011671)	0.080165** (0.032779)	0.057607 (0.037842)
No of obs.	892	911	465	465
No. of countries	120	121	59	59
Instruments	48	48	48	48
AR(2) (p)	0.359	0.177	0.660	0.549
Hansen (p)	0.509	0.602	0.451	0.401

The dependent variable is the HDI. Robust standard errors in parentheses. All estimations include time dummies (coefficients not reported). The asterisks (*, **, and ***) denote 10%, 5%, and 1% significance levels, respectively

specifically, the coefficients for mobile broadband in the developing country sample remained statistically significant, and the magnitude of the coefficient increased slightly to 0.000080. In the baseline regression of the developed country sample, the coefficient for internet bandwidth was 0.000099 and statistically significant at the 5% level. In this first robustness check, the coefficient increased slightly to 0.000126 and remained statistically significant at the 5% level.

To check if the results are robust to initial conditions, the system GMM using first differencing is applied as suggested by (Blundell & Bond, 1998). Estimated results for the developing countries sample are reported in Table 5, columns 1 and 2, and are found to be robust to the baseline estimation, except for an increase in the significance level of mobile broadband from 5 to 1% and the magnitude of the coefficient

to 0.000105. Regarding the developed countries sample, estimated results remain robust except for mobile broadband, column 3. Deviating from the baseline model, where the coefficient on mobile broadband is negative and statistically insignificant, it becomes significant at the 10% level. However, the transformation of the forward orthogonal deviations used in the initial specification tends to be superior, so we consider it for further analysis.¹⁵

For a further robustness test, we refer to the use of a two-step system GMM estimator, which in general provides more efficient estimates than a one-step GMM estimator (Blundell & Bond, 1998). Nonetheless, the two-step system GMM estimates sometimes converge to asymptotic distribution slowly, so the efficiency gain only sometimes materialises (Hwang & Sun, 2018). For this reason, a one-step system GMM estimation is conducted in which homoscedasticity and independence of the error term are assumed. Estimated results of the ICT variables stay robust to changes in the initial assumption, with one exception. In the developed country sample, mobile broadband becomes significantly negative at the 10% level (the results are not reported but are available upon request).

Results of the HDI Dimensions

Following Eq. (1), the estimation of the individual HDI dimensions as dependent variables is applied to investigate the channels through which ICT influences the overall index. Specifically, we test the relationship between the ICT variables on health, measured by life expectancy at birth, education, measured by expected years of schooling and mean years of schooling, and income, measured by GNI per capita (PPP \$). Data are obtained from the UNDP. For the estimation, identical specifications of the GMM are employed as those utilised in the baseline regressions. The results for the developing country sample are presented in Table 6.

The estimation results in columns 1, 3, and 5 demonstrate that mobile broadband was statistically significant at the 5% level for both health (0.000054) and education (0.000130). These findings suggest that a unit increase in mobile broadband is associated with a 0.000054 unit increase in the health index and a 0.000130 unit increase in the education index, holding all other variables constant. For internet bandwidth, none of the coefficients is statistically significant. These results indicate that not all dimensions of the HDI are affected equally by mobile broadband.

Table 7 presents the estimated results on the individual HDI dimensions for the developed countries sample. While the effects of mobile broadband on health and income, columns 1 and 5, are statistically insignificant, the impact of mobile broadband (−0.000150) on education, column 3, is statistically significantly negative at the 10% level. For internet bandwidth, the estimated coefficients for regressions on health and education in columns 2 and 4 are positive but statistically insignificant. However, internet bandwidth is statistically significant and positive at the 5% level on income (0.000122), as shown in column 6. In reference to the baseline results,

¹⁵ See, for example (Hayakawa, 2009) for an empirical Monte Carlo proof for the advantage of forward orthogonal deviation over first difference.

Table 6 Results HDI dimensions (developing countries)

Dependent variable	(1) Health	(2) Health	(3) Education	(4) Education	(5) Income	(6) Income
Health (t-1)	0.943852*** (0.013064)	0.949612*** (0.010233)				
Education (t-1)			0.947425*** (0.023865)	0.908215*** (0.042358)		
Income (t-1)					0.975140*** (0.017395)	0.873190*** (0.117499)
Mobile broad-band	0.000054*** (0.000020)		0.000130** (0.000053)		0.000053 (0.000047)	
Internet band-width		0.000379 (0.000348)		-0.000124 (0.000337)		0.000510 (0.002139)
GDP growth	-0.000067 (0.000069)	-0.000057 (0.000036)	0.000233 (0.000397)	0.000028 (0.000221)		
Pop. growth	-0.001958 (0.001243)	-0.001451 (0.001325)	-0.009937*** (0.003160)	-0.010887*** (0.002605)	-0.001720 (0.002366)	-0.006488 (0.006012)
Urban pop. growth	0.001136 (0.000930)	0.001553 (0.001300)	0.003418 (0.002407)	0.006088** (0.002523)	0.000689 (0.002282)	0.006744 (0.006820)
Inflation	0.000009 (0.000024)	0.000038 (0.000036)	0.000087 (0.000144)	-0.000000 (0.000124)	-0.000242 (0.000165)	-0.000277 (0.000211)
FDI	-0.000031 (0.000062)	-0.000029 (0.000052)	0.000089 (0.000316)	-0.000095 (0.000281)	0.000237 (0.000149)	0.000265 (0.000161)
Corruption	0.002675** (0.001284)	0.002640 (0.002206)	0.001755 (0.004635)	0.005460 (0.008352)	0.000786 (0.002975)	0.009628 (0.007676)
Constant	0.047300*** (0.011098)	0.042675*** (0.008785)	0.038273** (0.014846)	0.062063** (0.025495)	0.019152 (0.012366)	0.077665 (0.067463)
No of obs.	893	912	892	911	893	912
No. of countries	120	121	120	121	120	121
Instruments	48	48	48	48	43	43
AR(2) (p)	0.000406	0.000258	0.617	0.357	0.123	0.161
Hansen (p)	0.424	0.182	0.478	0.295	0.342	0.511

The dependent variables are the individual HDI dimensions. Robust standard errors are in parentheses. All estimations include time dummies (coefficients not reported). The asterisks (*, **, and ***) denote 10%, 5%, and 1% significance levels, respectively. The regression with health as the dependent variable in columns 1 and 2 does not pass the AR(2) test. Thus, the regression is tested directly on the dependent variable life expectancy at birth, out of which the health index of the HDI is created. The results confirm the main variable of interest and pass the AR(2) test. Results are available upon request. For the regression with income as the dependent variable in columns 5 and 6, the explanatory variable GDP growth is omitted due to high multicollinearity

Table 7 Results HDI dimensions (developed countries)

Dependent variable	(1) Health	(2) Health	(3) Education	(4) Education	(5) Income	(6) Income
Health (t-1)	1.005925*** (0.022060)	1.012291*** (0.022427)				
Education (t-1)			0.977094*** (0.032044)	0.997736*** (0.036482)		
Income (t-1)					0.940161*** (0.048305)	0.970432*** (0.024693)
Mobile broad-band	-0.000002 (0.000020)		-0.000150* (0.000079)		0.000079 (0.000068)	
Internet band-width		0.000024 (0.000037)		0.000115 (0.000078)		0.000122** (0.000052)
GDP growth	0.000035 (0.000084)	0.000048 (0.000063)	-0.000110 (0.000540)	-0.000107 (0.000453)		
Pop. growth	-0.001220* (0.000644)	-0.001002 (0.000638)	-0.000513 (0.003264)	-0.000905 (0.002881)	-0.001903 (0.002033)	-0.002833 (0.001750)
Urban pop. growth	0.000688 (0.000575)	0.000532 (0.000621)	0.001529 (0.002991)	0.002181 (0.002207)	0.003021 (0.001810)	0.003454* (0.001869)
Inflation	0.000010 (0.000037)	0.000019 (0.000029)	0.000128 (0.000133)	0.000144 (0.000161)	0.000049 (0.000074)	-0.000014 (0.000065)
FDI	0.000016* (0.000008)	0.000017** (0.000008)	-0.000088 (0.000077)	-0.000061 (0.000063)	-0.000014 (0.000038)	-0.000004 (0.000048)
Corruption	-0.003643*** (0.001343)	-0.003797*** (0.001408)	0.002852 (0.003740)	0.002492 (0.004015)	-0.001993 (0.003810)	-0.002461 (0.002857)
Constant	0.001300 (0.019272)	-0.004515 (0.019049)	0.032705 (0.027414)	0.002911 (0.028095)	0.049485 (0.035413)	0.030067 (0.020202)
No of obs.	465	465	465	465	465	465
No. of countries	59	59	59	59	59	59
Instruments	48	48	48	48	43	43
AR(2) (p)	0.0254	0.0215	0.950	0.887	0.507	0.434
Hansen (p)	0.531	0.691	0.284	0.478	0.387	0.248

The dependent variables are the individual HDI dimensions. Robust standard errors are in parentheses. All estimations include time dummies (coefficients not reported). The asterisk (*, **, and ***) denote 10%, 5%, and 1% significance levels, respectively. The regressions with health as the dependent variable in columns 1 and 2 do not pass the AR(2) test. Thus, the regression is tested directly on the dependent variable life expectancy at birth, out of which the health index of the HDI is created and by using an adjusted lag structure. The results confirm the main variable of interest finding and pass the AR(2) test. Results are available upon request. For the regression with income as the dependent variable in columns 5 and 6, the explanatory variable GDP growth is omitted due to high multicollinearity

internet bandwidth fosters human development mainly through the positive effect on income. In contrast, the negative effect of mobile broadband on education is not strong enough to impact the HDI as a composite index.

Discussion and Conclusions

This paper investigated the importance of ICT (represented by mobile and fixed telecommunication services) for human development (represented by HDI) in a panel of 121 developing and 59 developed countries from 2010–2017. We focused on this theme because previous studies on the relationship between ICT and human development missed it so far to account for the fast-changing technological environment, to use a large sample of developing and developed countries, and to consider the channels through which ICT influences the overall HDI. The main assumptions that developing and developed countries benefit differently depending on the respective telecommunication service and that only the new and more contemporary telecommunication services, mobile broadband, and internet bandwidth, are relevant are confirmed using various regressions. These findings are consistent with previous studies investigating this distinction in an earlier period (e.g. Lee et al., 2017). It is also consistent with one of the capabilities approaches' basic assumptions that the mere availability of resources, in this study, telecommunication services, does not automatically lead to the capability expansion of every individual.

The regressions for mobile telecommunications indicate that only mobile broadband has a statistically significant positive impact on human development, and only in developing countries. The results stay robust after employing various robustness tests, including sample adjustments, outlier tests, and changing model specifications. This finding is reasonable considering the leapfrogging hypothesis and the fact that mobile phones are a substitute for fixed lines only in developing countries. Another explanation might be diminishing returns to technology. While developed countries have reached saturation levels, developing countries have recently experienced relevant growth in mobile broadband. This simultaneously explains why mobile broadband and not mobile cellular subscriptions (mobile basic) have a statistically significant positive impact. Traditional mobile telecommunication services have become commonplace even in many developing countries and are not a particularly relevant driver of human development. Competitive advantages are gained using newer communication technologies, such as mobile broadband. This finding mainly extends upon previous literature that reveals a positive impact of mobile telecommunications on human development in developing countries (see Iqbal et al., 2019; Njoh, 2018; Lee et al., 2017) and indicates that the hitherto neglected indicator of mobile broadband is appropriate for future studies. Following the regression results of the individual HDI dimensions, improvements in health and education outcomes drive the positive effect of ICT on human development. Here, developing countries might benefit significantly from telemedicine which can improve maternal and child health outcomes through various interventions such as text messaging for health reminders, prenatal care, and vaccination reminders. In addition, enabling distance learning, supporting informal learning, and facilitating communication and collaboration in the learning environment are potential mechanisms through which mobile telecommunications positively impacts educational outcomes.

The competitive advantage argument also explains the statistically insignificant impact of fixed broadband subscriptions (broadband basic) in developed countries. The mere provision of broadband services is essential without significantly impacting human development. However, a higher quality of service, measured in terms of internet bandwidth, has a statistically significant positive impact on human development in developed countries. In this context, the increasing internet bandwidth through next-generation broadband can improve and provide unparalleled internet services such as cloud computing and high-quality real-time collaboration tools, thus benefiting consumers, businesses, and the economy. This result builds on previous literature showing a positive impact of fixed broadband telecommunications, represented by broadband basic, on human development in developed countries (see Petrić et al., 2020; Lee et al., 2017). However, according to our empirical estimation, internet bandwidth is the relevant proxy for fixed broadband telecommunications—a result especially relevant for future studies. As the regression results of the individual HDI dimensions show, the improvements in human development are gained exclusively through the positive effects on income. In this context, internet bandwidth potentially impacts income by facilitating the growth of the digital economy, fostering innovation and entrepreneurship, improving productivity, and increasing access to information and markets.

In developing countries, the lack of absorptive capacities enabling economies to gain from technological progress might explain the insignificant impact of fixed broadband telecommunications. In addition, two further explanations are relevant. First, the literature provides evidence for the critical mass phenomenon, according to which ICT diffusion must reach a certain threshold to benefit a country (Koutroumpis, 2009). In the present sample, the average number of broadband connections is around four per 100 inhabitants in developing countries. This relatively low number of users might be insufficient to benefit significantly from this technology. Second, while mobile phones overtook fixed phones in developing countries, mobile broadband potentially follows this path in overtaking fixed broadband, at least for fewer data-demanding services. In Africa, for example, broadband connections are based on extant fixed phone cables, which in their current state cannot compete with the availability and speed of mobile broadband offerings. Supposing there is no provision of new broadband technologies such as VDSL or fibre to the home, mobile broadband remains the preferred internet access method in many areas.

From a policy perspective, our study helps practitioners and policy makers to better understand the importance of considering the appropriate technology to improve the quality of life. Based on the conclusions, we recommend that developing countries prioritise the expansion of mobile broadband as a critical component of their ICT strategy to enhance existing growth effects. Here, the goal of nationwide coverage of affordable and reliable mobile broadband services is about more than just catching up with developments. It is about leapfrogging development stages and creating previously inconceivable use cases for improving health, education, and income. At an organisational level, these issues can be addressed by existing organisations investing in digitalisation and modernisation or by the creation of new innovative organisations. In this context, the government can provide specific support by granting financial incentives (e.g. subsidies, promotional loans), fostering an innovation-friendly environment (e.g.

reduction of bureaucracy, education initiatives), and opening markets (e.g. privatisation of public sectors, public-private partnerships). As fixed telecommunication services do not yet significantly impact developing countries, this study secondly recommends that governments develop a long-term broadband strategy and vision. Such a long-term perspective is particularly relevant since expanding the necessary infrastructure and adapting the population and companies require significantly more resources than the spread of mobile telecommunications. In many instances, effective governmental strategies encompass mechanisms that enable substantial engagement with the private sector, consumers, and pertinent stakeholders. A notable illustration is Chile, which was leading the way in implementing a national broadband strategy in Latin America. To foster economic expansion, Chile has authorised four regional providers to employ Worldwide Interoperability for Microwave Access (WiMAX) technology to accelerate broadband diffusion. Complementing this supply-side initiative, the demand-oriented strategy has incorporated initiatives aimed at enhancing digital literacy, e-government services, and general ICT diffusion. Consequently, most tax submissions occur through electronic means, and the implementation of e-procurement led to a more than two-fold increase in the number of transactions handled from 2005–2008 (Kelly & Rosotto, 2011). Finally, in developed countries, where basic fixed broadband networks are already widely available, governments should prioritise enhancing the quality of infrastructure. The provision of next-generation broadband can create access to various new opportunities and thus promote a lasting positive impact on human development. One promising approach to achieving this goal is to promote the development of competition. Competition in broadband supply is not only crucial for improving quality in terms of broadband speed and reliability but can also lead to increasing affordability and better customer service. Therefore, governments should place a priority on developing enabling policies that will facilitate competition throughout the supply chain, including international connectivity, domestic backbone, metropolitan connection, and local connection (World Bank, 2009).

The capability approach asserts that differences in human development exist not only between different groups of countries but also between regions, groups within regions, individuals within groups, and across various capabilities (channels). We have addressed this issue at the macro-level by examining which technology is developmental for which group of countries. Considering that this study utilised a large sample of developing and developed countries, future research could use a more disaggregated sample to examine more homogeneous groups of countries could yield valuable insights. Likewise, it is crucial to prioritise individual HDI dimensions and place them at the core of the analysis to fully comprehend the mechanisms of impact and offer specific policy recommendations. This is particularly relevant for the less researched education and health dimensions compared to income. Finally, insight into the impact of mobile broadband and internet bandwidth might be possible as more data becomes available. Interesting indicators include the actual use of mobile broadband, the sharing of mobile phones, and the actual average upload and download speeds experienced by users in a given country. We highlight these as promising topics for further research.

Appendix

Table 8 List of countries and their classification

Afghanistan	Dominica	Lao P.D.R.	Rwanda
Albania	Dominican Rep.	Latvia*	Saint Kitts and Nevis
Algeria	Ecuador	Lebanon	Saint Lucia
Angola	Egypt	Lesotho	St. Vincent and the Grenadines
Antigua and Barbuda	El Salvador	Liberia	Samoa
Argentina*	Equatorial Guinea	Libya	Sao Tome and Principe
Armenia	Eritrea	Lithuania*	Saudi Arabia*
Australia*	Estonia*	Luxembourg*	Senegal
Austria*	Eswatini	Madagascar	Serbia
Azerbaijan	Ethiopia	Malawi	Seychelles*
Bahamas*	Fiji	Malaysia*	Sierra Leone
Bahrain*	Finland*	Maldives	Singapore*
Bangladesh	France*	Mali	Slovakia*
Barbados*	Gabon	Malta*	Slovenia*
Belarus*	Gambia	Mauritania	Solomon Islands
Belgium*	Georgia	Mauritius	South Africa
Belize	Germany*	Mexico	South Sudan
Benin	Ghana	Micronesia	Spain*
Bhutan	Greece*	Moldova	Sri Lanka
Bolivia (Plurinational State of)	Grenada	Mongolia	Sudan
Bosnia and Herzegovina	Guatemala	Montenegro*	Suriname
Botswana	Guinea	Morocco	Sweden*
Brazil	Guinea-Bissau	Mozambique	Switzerland*
Brunei Darussalam*	Guyana	Myanmar	Tajikistan
Bulgaria*	Haiti	Namibia	Tanzania
Burkina Faso	Honduras	Nepal (Republic of)	Thailand
Burundi	Hong Kong, China*	Netherlands*	Togo
Cabo Verde	Hungary*	New Zealand*	Tonga
Cambodia	Iceland*	Nicaragua	Trinidad and Tobago
Cameroon	India	Niger	Tunisia
Canada*	Indonesia	Nigeria	Turkey*
Central African Rep.	Iran (Islamic Republic of)	North Macedonia	Turkmenistan
Chad	Iraq	Norway*	Uganda
Chile*	Ireland*	Oman*	Ukraine
China	Israel*	Pakistan	United Arab Emirates*
Colombia	Italy*	Palau*	United Kingdom*
Comoros	Jamaica	Panama	United States*
Costa Rica	Japan*	Papua New Guinea	Uruguay*

Afghanistan	Dominica	Lao P.D.R.	Rwanda
Côte d'Ivoire	Jordan	Paraguay	Uzbekistan
Croatia*	Kazakhstan*	Peru	Vanuatu
Cyprus*	Kenya	Philippines	Venezuela
Czech Republic*	Kiribati	Poland*	Viet Nam
Dem. Rep. of the Congo	Korea (Rep. of)*	Portugal*	Yemen
Denmark*	Kuwait*	Qatar*	Zambia
Djibouti	Kyrgyzstan	Russian Federation*	Zimbabwe

Developed countries have a very high HDI based on UNDP's classification and are labelled with an asterisk (*). Developing countries are countries with an HDI classified as low, medium, and high Funding Open Access funding enabled and organized by Projekt DEAL.

Declarations

Conflict of Interest The authors declare no competing interests.

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